

Multiple Choice (10 marks)

1. Which types of functions grow the slowest?
 - (a) $O(\log e^N)$
 - (b) $O(\log N)$
 - (c) $O(\log \log N)$
 - (d) $O(N)$
2. Church's thesis equates the concept of "computable function" with those functions computable by, for example, Turing machines. Which of the following is true of Church's thesis?
 - (a) It was first proven by Alan Turing.
 - (b) It has not yet been proven, but finding a proof is a subject of active research.
 - (c) It can never be proven.
 - (d) It is now in doubt because of the advent of parallel computers.
3. Let $f(X) = \text{if } x = 1 \text{ then } 0 \text{ else } [x * f(x - 1) + x^{**}2]$. The value of $f(4)$ is
 - (a) 53
 - (b) 29
 - (c) 50
 - (d) 100
4. What is the output of `"abc"[::-1]` in Python 3?
 - (a) Error
 - (b) abc
 - (c) cba
 - (d) c
5. The boolean expression `a[i] == max || !(max != a[i])` can be simplified to
 - (a) `a[i] == max`
 - (b) `a[i] != max`
 - (c) `a[i] < max || a[i] > max`
 - (d) `FALSE`

True / False (10 marks)

Answer True or False

6. True or False: In Python 3, the element at index 1 of the list ['a', 'Chemistry', 0, 1] is 'Chemistry'.
7. True or False: Confidentiality is compromised when someone logs into a system using a stolen login and password.
8. True or False: Reference counting is not effective for reclaiming cyclic structures due to its inability to detect circular references.
9. True or False: The method 'ab'.sort() can be used to sort the characters in the string 'ab' in place.
10. True or False: Static allocation is sufficient for managing the memory of recursive procedures without any issues.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a python function to check whether a sequence of numbers has an increasing trend or not.
12. Write a function to check if the given tuple has any none value or not.

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